SECOND METHOD TO CHECK CONVERGENCE OR DIVERGENCE

One of the most fundamental test for checking convergence of series is **Comparison Test**.

Now the questions is, **"How to compare two complex numbers?"**

The reason is that we cannot compare two complex numbers. Similarly we cannot compare two series of complex numbers. We always take modulus for comparing two complex numbers or terms of the series.

COMPARISON TEST

Let $\sum_{n=1}^{\infty} M_n$ be a convergent series of **real non − negative terms**. If $\{z_n\}$ is a sequence of complex numbers and $|z_n| \leq M_n$ for all n, then $\sum_{n=1}^{\infty} z_n = \sum_{n=1}^{\infty} (x_n + i\,y_n)$ converges.

EXAMPLE:

Show that $\sum_{n=1}^{\infty} \frac{(3+4\,i)^n}{5^n\,n^2}$ converges.

SOLUTION:

Consider

$$|z_n| = \left|\frac{(3+4\,i)^n}{5^n\,n^2}\right| = \frac{|3+4\,i|^n}{5^n\,n^2} = \frac{\left(\sqrt{3^2+4^2}\right)^n}{5^n\,n^2} = \frac{\left(\sqrt{25}\right)^n}{5^n\,n^2} = \frac{5^n}{5^n\,n^2} = \frac{1}{n^2}$$

We know that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

So, by Comparison Test, $\sum_{n=1}^{\infty} \frac{(3+4\,i)^n}{5^n\,n^2}$ converges.

REMARK: The Comparison Test is also used for the series of real numbers.

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EXAMPLE 1: Show that the series

$$\sum_{n=0}^{\infty} \frac{(1-i)^n}{n!}$$

converges.

SOLUTION:

Given $\xi_n = \dfrac{(1-i)^n}{n!}$

Consider $$\lim_{n\to\infty} \frac{|\xi_{n+1}|}{|\xi_n|} = \lim_{n\to\infty} \frac{\left|\dfrac{(1-i)^{n+1}}{(n+1)!}\right|}{\left|\dfrac{(1-i)^n}{n!}\right|}$$

$$= \lim_{n\to\infty} \left|\frac{(1-i)^{n+1}}{(n+1)!}\right| \left|\frac{n!}{(1-i)^n}\right|$$

$$= \lim_{n\to\infty} \left|\frac{(1-i)}{n+1}\right|$$

$$= \lim_{n\to\infty} \frac{|1-i|}{n+1}$$

Since $|1-i| = \sqrt{1^2+(-1)^2} = \sqrt{2}$

$$\lim_{n\to\infty} \frac{|\xi_{n+1}|}{|\xi_n|} = \lim_{n\to\infty} \frac{\sqrt{2}}{n+1}$$

$$= \frac{\sqrt{2}}{\infty} = 0 < 1$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{(1-i)^n}{n!} \text{ converges.}$$

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$\Rightarrow u_{xx} + u_{yy} = 0$

Similarly we can prove that $v_{xx} + v_{yy} = 0$

Hence $u(x, y)$ and $v(x, y)$ are harmonic functions.

EXAMPLE 1: Consider the function $f(z) = z^2$. Show that its component functions are harmonic functions.

SOLUTION:

Given function is $f(z) = z^2$

As $z = x + iy$, so $f(x + iy) = (x + iy)^2$

$$= x^2 - y^2 + i(2xy)$$

Since $f(z) = z^2$ is a polynomial, so it is an entire function. Therefore, it is analytic for all z in complex plane. Hence, by theorem 1, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$ are harmonic functions.

EXAMPLE 2: Consider the function

$$f(z) = e^{-y} \sin x + i e^{-y} \cos x$$

Show that its component functions are harmonic functions.

SOLUTION:

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